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Arduino-based RF Controlled Robot

Arduino-based projects in EFY magazine are a welcome beginning. I am sure many like me who dabble with Arduino will look forward to inclusion of more Arduino-based projects in the future. The project 'Arduino-based RF Controlled Robot' published in EFY Dec. 2012 issue is excellent.

R.K. Bose
New Delhi

EFY: Thank you for such positive feedback. We have published Arduino projects in the past also, and shall try to include more projects on different Arduino boards in the future.

LED Voltmeter

The 'LED Voltmeter' circuit published in July 2012 issue is working as expected. It gave correct output when I simulated it using Multisim software to check proper working of the circuit. But when I assembled the circuit on a breadboard, the 3V and 6V LEDs glowed without any input. Could you please tell me where I have gone wrong?

Jelson Chacko
Pune

The author Raj K. Gorkhali replies:
Check proper power supply, it should

'Spot An Error' Award Winners

□ In 'Real-Time Clock' construction project published in EFY Dec. 2012, X1 component shown in Fig. 5 is for 12V AC input from the transformer's secondary terminals.

—Manicka Vasagam Sundaresan

□ In 'USB Charger for Mobile Phones' circuit, the equation for V_{OUT} is incorrect. It should be $V_{OUT} = 1.195(1 + (R2/R1))$

—Dr Ramkumar Ramaswamy

□ In 'USB Charger for Mobile Phones' circuit, the equation for V_{OUT} is incorrect. It should be $V_{OUT} = 1.195(1 + (R2/R1))$

—Jayesh Patil

be 12V. Also check proper values for resistors R2 and R3. Both should be 1 kilo-ohm each. Also, instead of two LEDs, try using a single LED at the output of each op-amplifier. I hope this will solve the problem.

1kW Sinewave Inverter

Regarding the '1kW Sinewave Inverter' construction project published in Nov. 2012 issue, I have following queries:

1. In the description, 'transistor' is written in place of 'MOSFET'.

2. On page 113, it is mentioned that "Two 2.2µF capacitors are connected to the gates of the MOSFETs." But where are the capacitors?

3. Details of the batteries (BATT.1 and BATT.2) are not given in the text.

4. The SCR operates in two ways ('on' and 'off'). How is it possible?

Anil Kumar Rout
Bose Engg College
Cuttack, Odisha

The author Dr R.V. Dhakale replies:

1. There is no mistake because two types of transistors are used in the circuit: metal-oxide semiconductor field-effect transistor (MOSFET) and bipolar transistor. Notations T1 through T8 are used for MOSFETs and notations T9 through T11 are used for bipolar transistors.

2. In the article, it is written that if sinewave is not obtained at the output of the inverter, a capacitor (2.2µF) can be connected to each bank with respect to ground. Options are given. Hence no such capacitors are shown in the circuit.

3. Batteries BATT.1 and BATT.2 are similar. They can be 12V, 7Ah to 12V, 180 Ah, depending on how long backup is required. If two batteries of 12V, 7Ah are used, backup duration will be about 10 minutes. But with 12V, 180Ah batteries, backup duration will be about four hours.

4. SCR is connected across the supply of IC CD4047. So when SCR is not conducting, the voltage across its anode and cathode is 5.1V. When SCR is conducting, the voltage across its anode and cathode is 0.7V. The same voltage goes to the IC. Hence it will stop producing 5V pulses at its pins 10 and 11, switching off the inverter.